

PHASE 5 – PROJECT DEVELOPMENT PHASE

Project Title: FitBuddy – AI Fitness Plan Generator using Gemini Models

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1. Development Overview

The development phase converts the approved FitBuddy design and requirements into a working application. The implementation should follow the workflow defined in Phase 3 and the tasks planned in Phase 4.

2. Development Modules

Module	Implementation Purpose
Input Form	Collect age, fitness goal, activity level, workout days and optional preferences.
Validation	Ensure required inputs are present and usable.
Prompt Builder	Combine user inputs into a clear prompt for the Gemini model.
Gemini Integration	Send the prompt to the Gemini service using the project configuration.
Response Handler	Receive the AI response and prepare it for display.
Plan Display	Show the generated schedule, exercises and general guidance in readable sections.

3. Development Workflow

- Enter user details.
- Click Generate Plan.
- Validate required inputs.
- Prepare the AI prompt.
- Send request to Gemini.
- Receive generated response.
- Display the structured plan.

4. Prompt Structure

A suitable project prompt can request Gemini to use the supplied age, fitness goal, activity level, workout days and optional preferences to produce a clear, structured fitness plan. The output should be organized into a schedule and exercise recommendations appropriate to the supplied information.

5. Error Handling

- Display a message when required inputs are missing.
- Handle API/network failures with a user-friendly error message.
- Avoid exposing API keys or private configuration values.
- Handle empty or unusable AI responses gracefully.

6. Development Deliverables

- Application source code.

- Project README with setup and usage instructions.
- Configuration guidance for Gemini access.
- Screenshots of the implemented interface.
- Phase-wise project documents.
- Final working version for testing and demonstration.

7. Important Submission Note

Actual implementation details, screenshots, repository URL and deployed/live URL should be added after the team has completed the corresponding work. This document describes the development deliverables and workflow without inventing a completed implementation.

8. Development Conclusion

The development phase produces the FitBuddy application according to the design and requirements. The completed application then moves to systematic testing.